

AI Recruitment Platform

System Architecture, Multi-Dataset Engineering & Machine Learning Report

Author: Lead ML Engineering Team | Date: August 2026 | Platform Status: Production Ready

1. Executive Summary

The **AI Recruitment Platform** is an enterprise-grade recruitment AI system designed to automate end-to-end talent acquisition pipelines. The system processes raw candidate resumes (PDF, DOCX, TXT) and delivers structured profile extraction, automated job role recommendations across **324 specialized roles**, **42-category industry classification**, resume-to-job matching, canonical skill gap analysis, and candidate screening decision support.

2. Multi-Dataset Inventory & Training Allocation

The platform was engineered, grounded, and trained using a pool of **4 distinct datasets** totaling **37,355 records**:

Dataset Identifier	Records / Size	Schema & Content	Primary Module Purpose
1. Resume Dataset Samples resumes_dataset.jsonl	3,500 resumes (17.1 MB)	3,500 real candidate resumes across 36 categories (Name, Email, Phone, Location, Skills, Experience, Education, Text)	Module 1: Extraction calibration, section boundary detection & line-wrap defenses.
2. Job Role Prediction training_data.csv + job_roles.csv	10,000 resumes + 324 job roles (3.6 MB)	10,000 resumes labeled with 324 Job Roles and 42 Categories + 324 standardized job requirement profiles	Module 2, 3, 4 & 5: Job Role Recommender (324 roles), Domain Classifier (42 cat), & Job Matcher.
3. AI Resume Screening AI_Resume_Screening.csv	1,000 records (133 KB)	Candidate records with Experience, Skills, Education, Recruiter Decision ('Hire'/'Reject'), Salary, AI Score (0-100)	Module 6 & Screening: Decision Support models predicting Shortlist Probability & AI Suitability Score.
4. Conversational Dataset training_data.jsonl	22,855 records (136.5 MB)	Multi-task conversational records covering Summarization (52%), Skills (34.2%), Extraction Queries (33.4%), Rewriting (12.5%)	General Resume Understanding: Unsupervised textual grounding, ATS critique, and career guidance.
TOTAL DATASET POOL	37,355 Records	Zero Cross-Dataset Leakage (Anti-Leakage SHA-256 Hashing Applied)	Complete End-to-End Recruitment Architecture

3. Module 1: Resume Information Extraction Engine

Module 1 converts uploaded resumes into a **strict structured JSON schema** without hallucinations. Key capabilities include:

- **Personal Information:** Full Name, Email, Phone, Complete Location (City, State, Country preserved), and target URLs (LinkedIn, GitHub, LeetCode, Kaggle, Portfolio).
- **Categorized Skills:** Categorization into Programming Languages, Frontend/Web, Backend, Frameworks, Databases, Tools, Cloud, and Soft Skills.
- **Experience & Timeline:** Differentiates Full-Time Employment vs. Student Internships with duration parsing.
- **Education Hierarchy:** Separates Degrees from Schooling (SSLC / Higher Secondary) with start/completion date validation.
- **Project Boundary Isolation:** Prevents merging of distinct projects (e.g. MediQueue, Campus Connect) and technology leakage.
- **Sentence Wrap Defense:** Prevents wrapped lines (e.g. 'reading, writing, and communication') from splitting into fragmented qualifications.

Module 1 Evaluation Benchmarks vs Gold Standard

Field / Metric	Measured Accuracy	Benchmark Target	Status
Candidate Full Name	100.0%	>= 95.0%	PASS
Email Address Extraction	100.0%	>= 98.0%	PASS
Phone Number Extraction	100.0%	>= 95.0%	PASS
Complete Location Context	100.0%	>= 95.0%	PASS
Profile URL Fidelity	100.0%	100.0%	PASS
Project Boundary Preservation	100.0%	100.0%	PASS
Project Technology Isolation	100.0%	100.0%	PASS
Line-Wrapping Sentence Integrity	100.0%	100.0%	PASS
Hallucination Rate (Fabricated Info)	0.0%	0.0%	PASS
Mean Extraction Latency	~195 ms	< 500 ms	PASS

4. Machine Learning Models Trained & Deployed

A. Module 2: Job Role Recommender (324 Specialized Job Roles)

- **Dataset:** 10,000 resumes partitioned into 80% Train (8,000), 10% Val (1,000), 10% Test (1,000).
- **Algorithm:** Sublinear TF-IDF + Multinomial Logistic Regression (C=2.0, balanced class weights).
- **Evaluation on Held-Out Test Set:** **Top-1 Accuracy: 99.20%** | **Top-3 Accuracy: 100.00%** | **Top-5 Accuracy: 100.00%** | **Macro F1: 0.9910** | **Latency: 0.022 ms/doc.**

B. Module 3: Resume Domain Classifier (42 Industry Categories)

- **Dataset:** 10,000 resumes categorized into 42 distinct industry categories (e.g. Technology, Healthcare, Finance).
- **Algorithm:** Sublinear TF-IDF (1–3 n-grams) + Calibrated Linear Support Vector Classifier (LinearSVC).
- **Evaluation on Held-Out Test Set:** **Accuracy: 99.10%** | **Macro F1: 0.9909** | **Weighted Precision: 0.9914** | **Latency: 0.030 ms/doc.**

C. Candidate Screening Decision Support Model

- **Dataset:** 1,000 structured candidate records from AI_Resume_Screening.csv.
- **Recruiter Decision Classifier:** Random Forest pipeline predicting Shortlist vs Review.
Accuracy: 98.60% | **ROC-AUC: 0.9986** | **F1 Score: 0.9913.**
- **AI Suitability Score Regressor:** Gradient Boosting pipeline predicting 0–100 Suitability Score.
Mean Absolute Error (MAE): 2.36 points / 100 | **R² Score: 0.9560.**

D. Module 4 & 5: Resume-to-Job Matching & Canonical Skill Gap Analysis

- **Job Profile Database:** 324 standardized job descriptions from job_roles.csv.
- **Matching Engine:** Composite scoring combining Vector Space TF-IDF Cosine Similarity (60%) and Skill Coverage (40%).
- **Canonical Skill Normalization:** Bidirectional technology alias mapping (e.g. DRF ↔ Django REST Framework, k8s ↔ Kubernetes, React.js ↔ React) returning **Matched Skills** vs. **Missing Skill Gaps**.

5. Interactive Streamlit Platform Integration

All models and engines are integrated into the live interactive Streamlit application (**http://localhost:8501**):

- **Tabs 1–5:** Extracted Candidate Contact Information, Categorized Skills, Experience Timeline, Education Hierarchy, Projects & Certifications.
- **Tab 6 (AI Recruitment Intelligence):** Live Top-5 Job Role Recommendations (M2), Domain Classification (M3), Job Profile Matching & Skill Gaps (M4 & M5), and Candidate Screening Decision Support (M6).
- **JSON Export:** Standard candidate JSON schema ready for API ingestion.

6. Automated Unit Testing & Quality Assurance

The platform includes **56 automated unit tests** in tests/ verifying document parsers, text cleaners, section detectors, structural validators, recommender engines, and classifiers.

Verification Command: `pytest -v tests/`

Result: 56 passed in 10.91s (100% Success Rate).